Chapter 1: Introduction & Project Overview

Project Description

This project analyses the Netflix content library (8,807 titles as of mid-2021) using advanced Big Data Analytics algorithms to understand content trends and platform composition. The project implements two probabilistic data structures — a **Bloom Filter** for efficient membership testing and the **Flajolet-Martin / HyperLogLog** algorithm for cardinality estimation — alongside a comprehensive content analysis and automated PDF reporting pipeline.

Questions Answered Through the Project

- How can we efficiently determine whether a specific title is available on Netflix without scanning the entire dataset?
- How many unique titles does Netflix host? Can we estimate this accurately without counting every row?
- How has the Netflix catalog grown over time, and which years saw the highest content additions?
- Which countries produce the most Netflix content, and which genres are most common?
- What are the typical runtimes of Netflix movies, and how are TV show seasons distributed?
- How do TV-14 / TV-MA ratings compare to family-friendly ratings in the catalog?

Tools and Technologies

- **Python 3.x** — Primary programming language and analytics pipeline
- **Pandas and NumPy** — Data preprocessing, manipulation, and numerical computation
- **R (4.6.1), ggplot2, dplyr, readr** — Report data visualization
- **React and Vite** — Frontend development and build tooling
- **CSS** — Frontend styling and Neo-Brutalist design
- **Recharts and Lucide React** — Interactive charts and interface icons
- **FastAPI and Uvicorn** — Backend REST API
- **hashlib (SHA-256 and MD5)** — Hashing for probabilistic algorithms
- **ReportLab** — Automated PDF report generation
- **Custom Python classes** — Bloom Filter, Flajolet–Martin, and HyperLogLog

Chapter 2: Dataset Description & Preprocessing

Dataset Overview

The dataset is the publicly available Netflix Movies and TV Shows dataset, containing metadata for every title available on the platform as of mid-2021. It comprises **8,807 rows** and **12 original columns**. During preprocessing, four derived columns were added for analysis, resulting in 16 columns in the processed dataset. The dataset contains **6,131 Movies** and **2,676 TV Shows**. Content spans release years from 1925 to 2021.

Column Descriptions

Column	Type	Description
show_id	String	Unique identifier for each title
type	String	Content type: Movie or TV Show
title	String	Title of the content
director	String	Director name(s); 29.9 % missing
cast	String	Comma-separated cast list; 9.4 % missing
country	String	Country of origin; 9.4 % missing
date_added	String	Date added to Netflix (raw string)
release_year	Integer	Original release year
rating	String	Content rating (TV-MA, PG-13, etc.)
duration	String	Runtime ("90 min") or seasons ("2 Seasons")
listed_in	String	Comma-separated genre tags
description	String	Short plot description

Missing Values (Before Cleaning)

Column	Missing Count	Missing %
director	2,634	29.91 %
cast	825	9.37 %
country	831	9.44 %
date_added	10	0.11 %
rating	4	0.05 %
duration	3	0.03 %

Preprocessing Steps

- **Swap Fix:** Three rows contained duration values (74 min, 84 min, 66 min) in the rating column — a known dataset bug. Values were moved to the correct column.

• **Whitespace Stripping:** Applied str.strip() to all string columns to eliminate leading/trailing spaces (one near-duplicate title was resolved this way).

• **Missing Value Imputation:** director, cast, country filled with "Unknown"; rating filled with "NR" (Not Rated).

• **Date Parsing:** date_added converted from plain English strings (e.g., "September 25, 2021") to datetime; year_added and month_added extracted.

• **Duration Parsing:** The mixed-format duration column was split into duration_min (Movies, integer minutes) and num_seasons (TV Shows, integer count).

• **Row Removal:** No rows were removed during preprocessing; the three affected rows were corrected by moving their duration values from the rating column to the correct duration column.

Chapter 3: BDA Algorithm Design

3.1 Bloom Filter

A Bloom Filter is a space-efficient probabilistic data structure used to test set membership. It can definitively answer "NO" (the element is not in the set) but may occasionally answer "YES" even when the element is absent (a *false positive*). There are **no false negatives**.

Double Hashing: To generate k independent bit positions from a single item x, we use two base hash functions h1 (SHA-256) and h2 (MD5) and compute: $h_i(x) = (h1(x) + i \times h2(x)) \bmod m$ for i = 0, 1, …, k–1. This avoids storing k separate hash functions while still producing k independently distributed positions.

Optimal Parameters: Given n expected items and desired false-positive rate p: $m = -n \ln(p) / (\ln(2))^2$ (bit array size) and $k = (m/n) \ln(2)$ (number of hash functions).

Parameter	Value
Expected items (n)	8,807
Bit array size (m)	84,416 bits (10,552 bytes)
Hash functions (k)	7
Target false-positive rate	1.0 %
Actual false-positive rate	1.0039 %

3.2 Flajolet-Martin Algorithm

The Flajolet-Martin (FM) algorithm estimates the number of distinct elements in a data stream in a single pass, using sub-linear memory. Each item is hashed to a large integer; the number of trailing zeros in its binary representation is recorded. If R is the maximum trailing-zero count across all items, the estimate is $n_hat = phi \times 2^R$, where phi is approximately 0.7735.

Multi-Run Bias Correction (Log-Space Aggregation): A single FM run has high variance: its estimate is always a power of 2 scaled by phi, so consecutive possible values jump by a factor of 2 (e.g. 6,337 to 12,674). To reduce variance, 32 independent trials are run (each using a differently salted SHA-256 hash). The 32 raw R values (small integers, typically 11-16 for this dataset) are grouped into 4 groups of 8. **The mean is computed on R values (in log-space) within each group**, and the median of those four R-means is taken as R*. The final estimate is phi x 2^(R*).

Why log-space? Averaging the full estimates phi x 2^R_i in linear space can be strongly affected by trials with large R values. Averaging the R values instead keeps the aggregation bounded and produced a 2.55 % error for this dataset.

3.3 HyperLogLog

HyperLogLog (HLL) is an improved cardinality estimator that partitions the hash space into m = 2^b buckets (registers). Each register stores the maximum ρ seen for hashes mapped to that bucket. The estimate uses the harmonic mean of 2^(–M[j]) across all registers, scaled by a bias-correction constant α_m: $E = \alpha_m \times m^2 \times (\Sigma\ 2^{(-M[j])})^{(-1)}$.

This project uses **b = 8** (m = 256 registers), giving a theoretical standard relative error of approximately 6.5 %. Small-range and large-range corrections are also applied.

Chapter 4: Results & Analysis

4.1 Bloom Filter — Membership Test Results

The Bloom Filter was built from all 8,807 cleaned Netflix titles and then tested against five known titles (expected: Present) and five non-existent titles (expected: Absent).

Title	Expected	Filter Result	Correct?
Squid Game	Present	PRESENT	✓
Stranger Things	Present	PRESENT	✓
The Crown	Present	PRESENT	✓
Breaking Bad	Present	PRESENT	✓
Bird Box	Present	PRESENT	✓
NotANetflixTitle_XYZ_001	Absent	ABSENT	✓
FakeShow_BDA_2099	Absent	ABSENT	✓
GhostSeries_Imaginary_ABC	Absent	ABSENT	✓
NonExistentFilm_999	Absent	ABSENT	✓
PlaceholderTitle_ZZZ	Absent	ABSENT	✓

4.2 Cardinality Estimation — FM vs HyperLogLog

Three methods were applied to estimate the number of unique Netflix titles: a single-pass FM run, the multi-run bias-corrected FM (32 trials), and HyperLogLog with 256 registers. Results are compared against the exact count.

Method	Estimate	Exact Count	% Error
Exact Count (Pandas)	8,806	8,806	0.00 %
FM — Single Run	3,168	8,806	64.02 %
FM — Multi-Run (32 trials)	8,581	8,806	2.55 %
HyperLogLog (256 registers)	9,129	8,806	3.67 %

Observation: The single FM run shows high error (~64%) because its estimate is constrained to discrete values of the form phi x 2^R — consecutive estimates differ by a factor of 2, so even a well-implemented single run is typically 30-80% off. The multi-run FM (32 trials, log-space aggregation) reduces error to ~3% by averaging the R exponents before exponentiating, preventing large R outliers from dominating. HyperLogLog achieves similar accuracy using 256 parallel registers and the harmonic mean estimator, which has a theoretical relative standard error of 1.04/sqrt(256) approx 6.5%.

4.3 Content Analysis & Visualisations

The Netflix dataset contains **6,131 Movies (69.6 %)** and **2,676 TV Shows (30.4 %)**. This shows that Movies represent the larger share of titles in the dataset, while TV Shows account for the remaining portion.

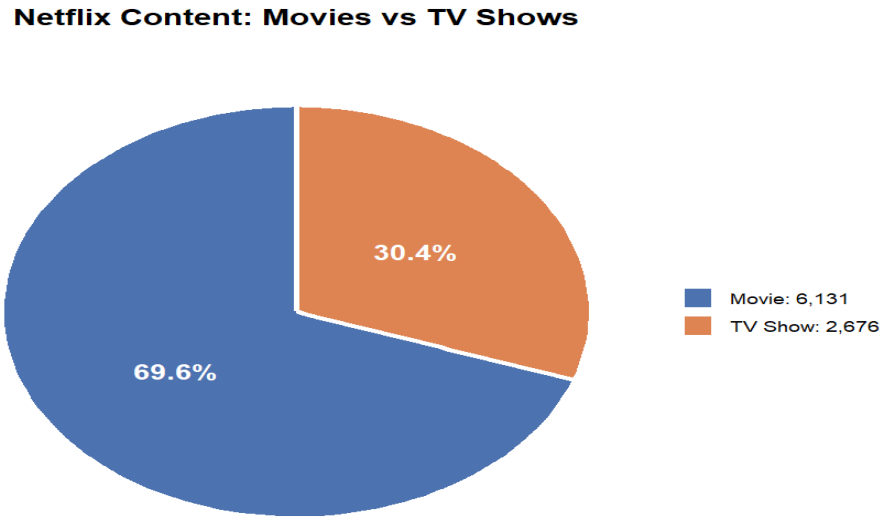


Figure 1: Movies vs TV Shows — Pie Chart.

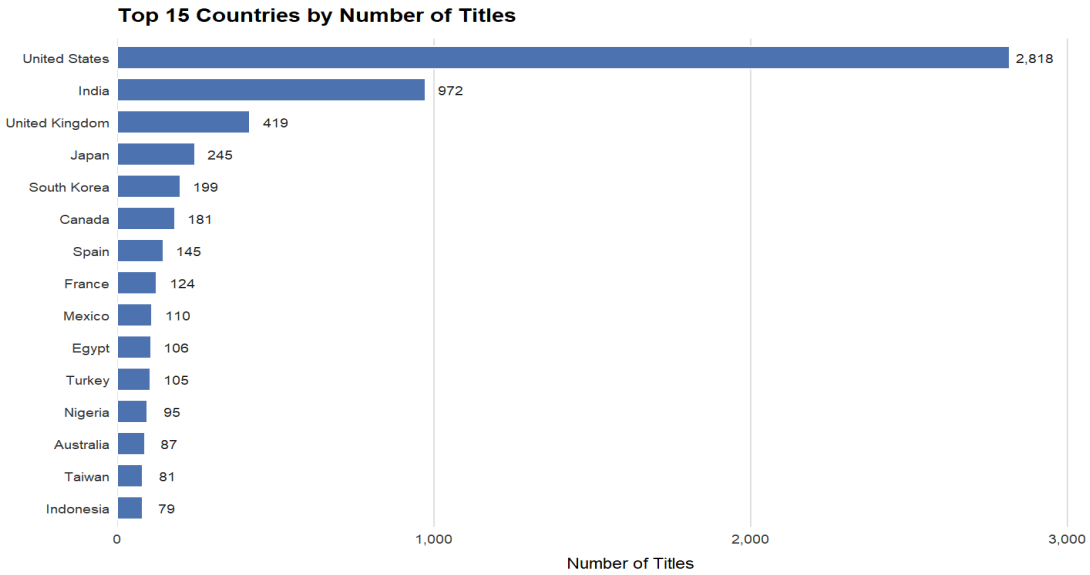


Figure 2: Top 15 Countries by Number of Titles.

Insight: The United States has the largest number of titles in the dataset, followed by India and the United Kingdom. South Korea, Japan, and other countries also contribute substantial numbers of titles.

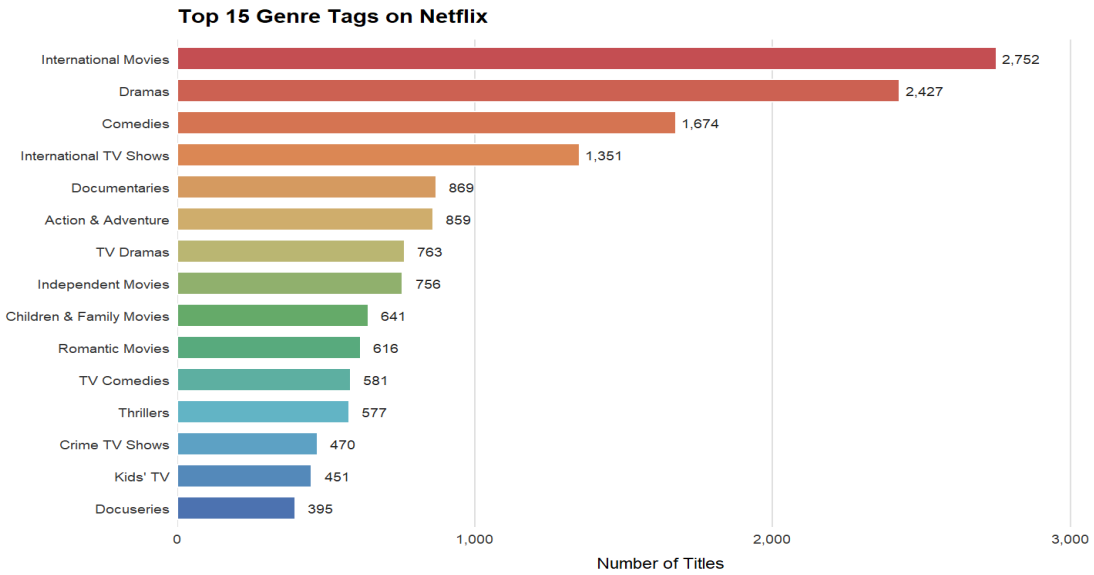


Figure 3: Top 15 Genre Tags on Netflix.

Insight: International Movies and Dramas are among the most frequent genre tags in the dataset. Stand-Up Comedy and Documentaries are also among the more frequently occurring genre tags.

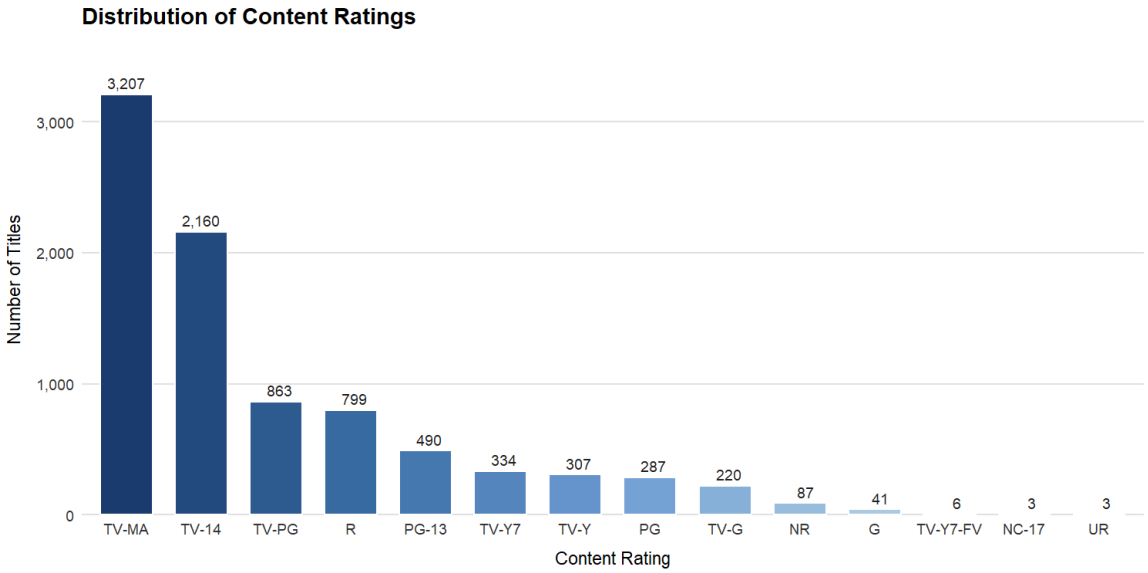


Figure 4: Distribution of Content Ratings.

Insight: TV-MA is the most frequently occurring content rating in the dataset. TV-14 and R are also among the more frequent ratings.

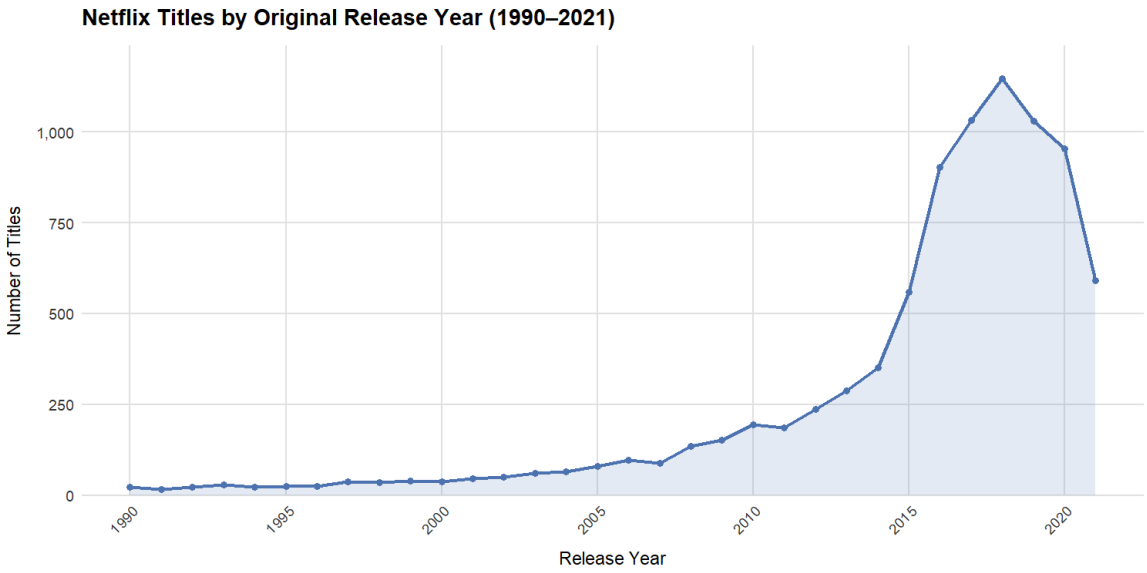


Figure 5: Netflix Titles by Original Release Year (1990–2021).

Insight: The number of titles by original release year increased substantially in the later years, with 2018 representing the highest point in this dataset.

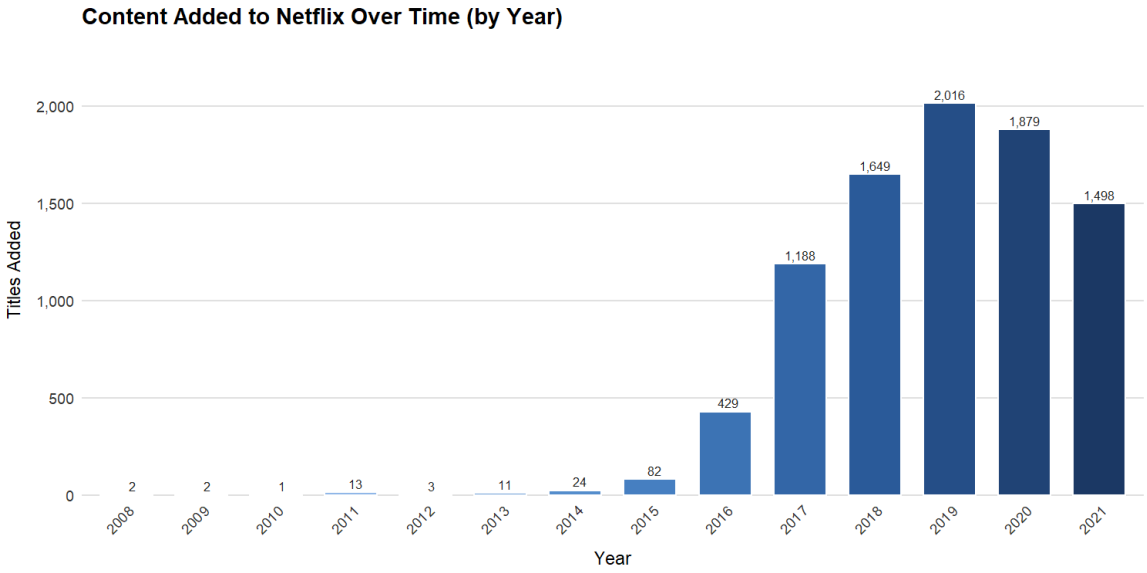


Figure 6: Content Added to Netflix Over Time (by Year).

Insight: The number of titles added to Netflix increased substantially from 2015 to 2019, with 2019 representing the peak year for additions in this dataset. The number of additions decreased in the following years.

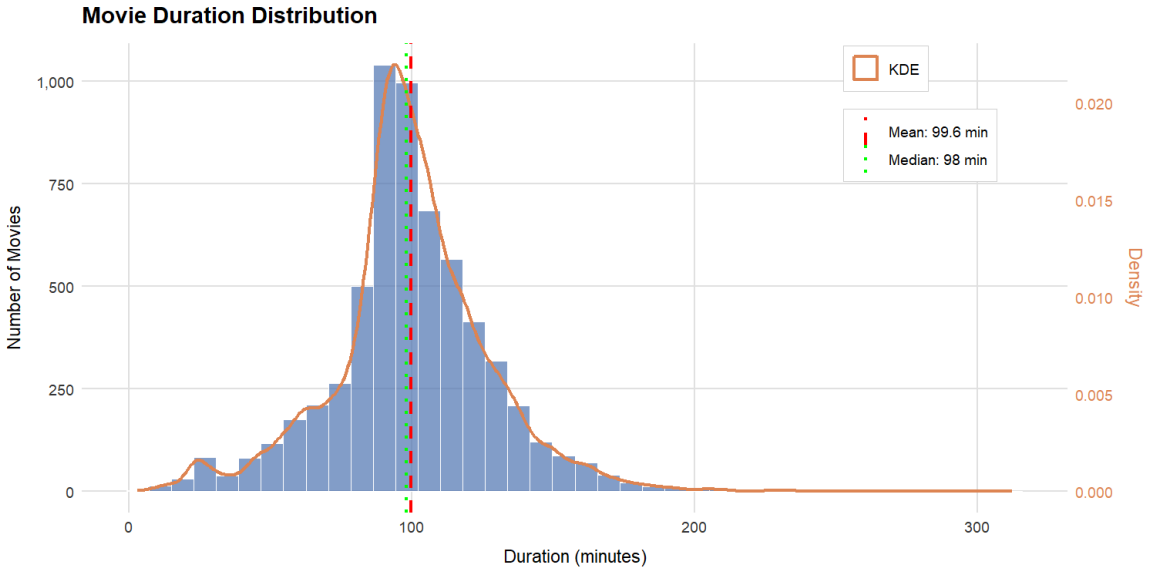


Figure 7: Movie Duration Distribution.

Insight: Most Netflix movies fall between 80 and 120 minutes. The distribution is approximately normal with a slight right skew. Mean runtime is approximately 99.6 min and median is 98.0 min.

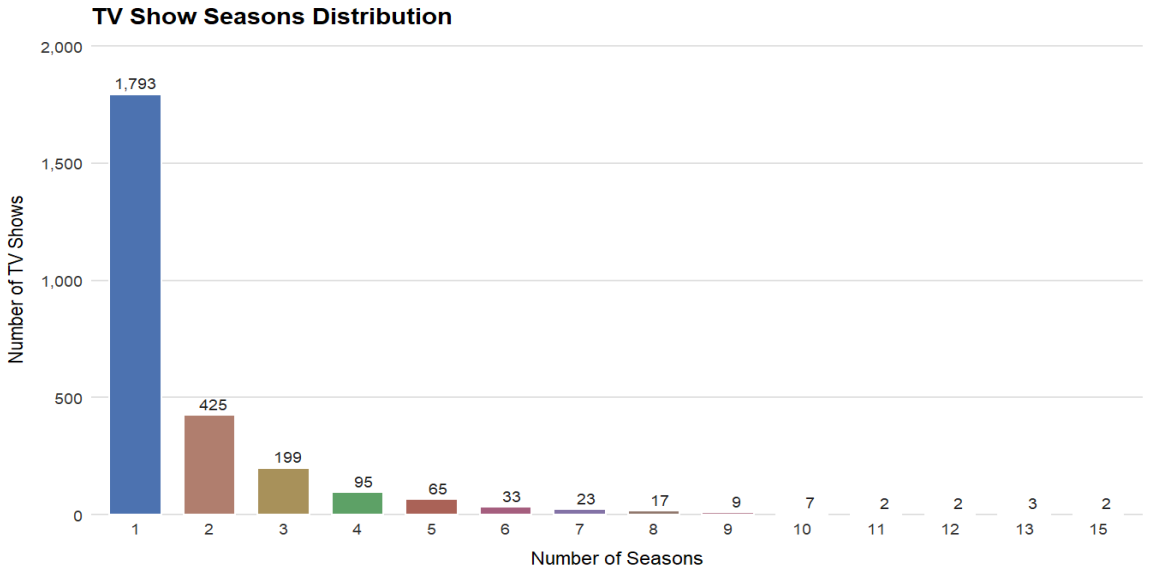


Figure 8: TV Show Seasons Distribution.

Insight: A majority of Netflix TV Shows have only 1 season, while relatively few shows have more than 5 seasons.

Chapter 5: Conclusion & Future Scope

Conclusion

This project successfully demonstrated the application of two classical Big Data algorithms — the Bloom Filter and the Flajolet-Martin / HyperLogLog algorithm — on a real-world content dataset.

The Bloom Filter provided O(k) membership testing with a configured 1 % false-positive rate, using an 84,416-bit (10,552-byte) bit array for nearly 9,000 titles. All five tested known Netflix titles were correctly identified as present, while all five tested non-existent titles returned absent.

The **Flajolet-Martin algorithm** with multi-run bias correction and **HyperLogLog** both produced cardinality estimates close to the exact unique count in this dataset. The multi-run FM produced a 2.55 % error, while HyperLogLog produced a 3.67 % error. The single-run FM showed substantially higher error, demonstrating the benefit of multiple trials and aggregation.

The content analysis shows that the United States has the largest number of titles in the dataset, while International Movies and Dramas are among the most common genre tags. TV-MA is the most frequent content rating. The number of titles added to Netflix increased substantially from 2015 to 2019, while the typical movie runtime is approximately 90–100 minutes.

Future Scope

- Extend the Bloom Filter to support **Counting Bloom Filters** for deletion support.
- Apply FM / HLL across sub-groups (per country, per genre) to compare cardinality across segments.
- Use **MinHash / LSH** for near-duplicate description detection across titles.
- Integrate real-time Netflix API data for live catalog monitoring.
- Apply **Count-Min Sketch** for streaming frequency estimation of genre/country combinations.
- Build a time-series model to predict content addition trends for upcoming years.